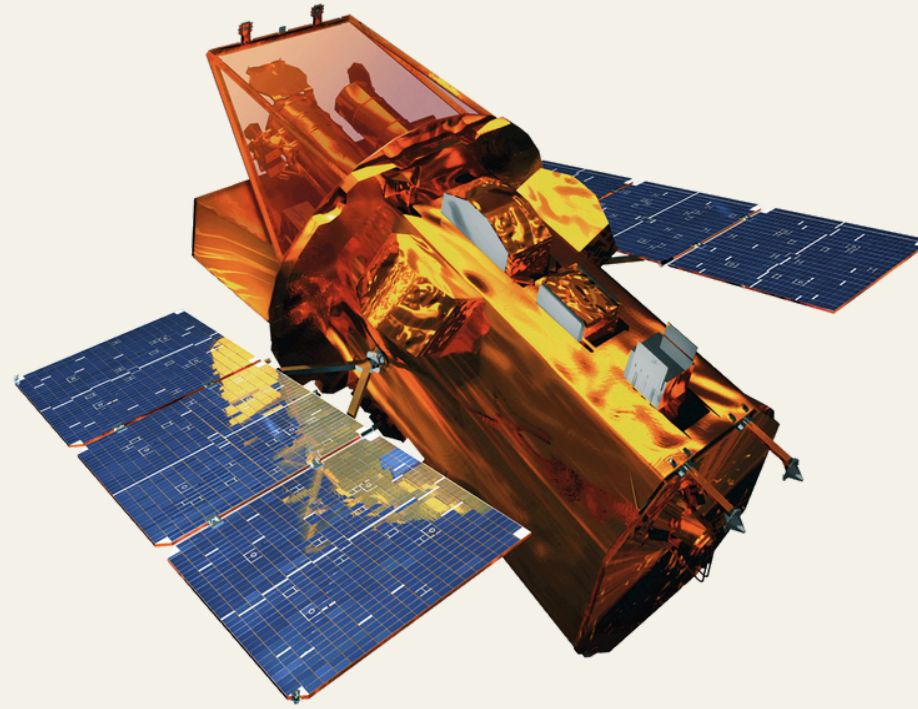
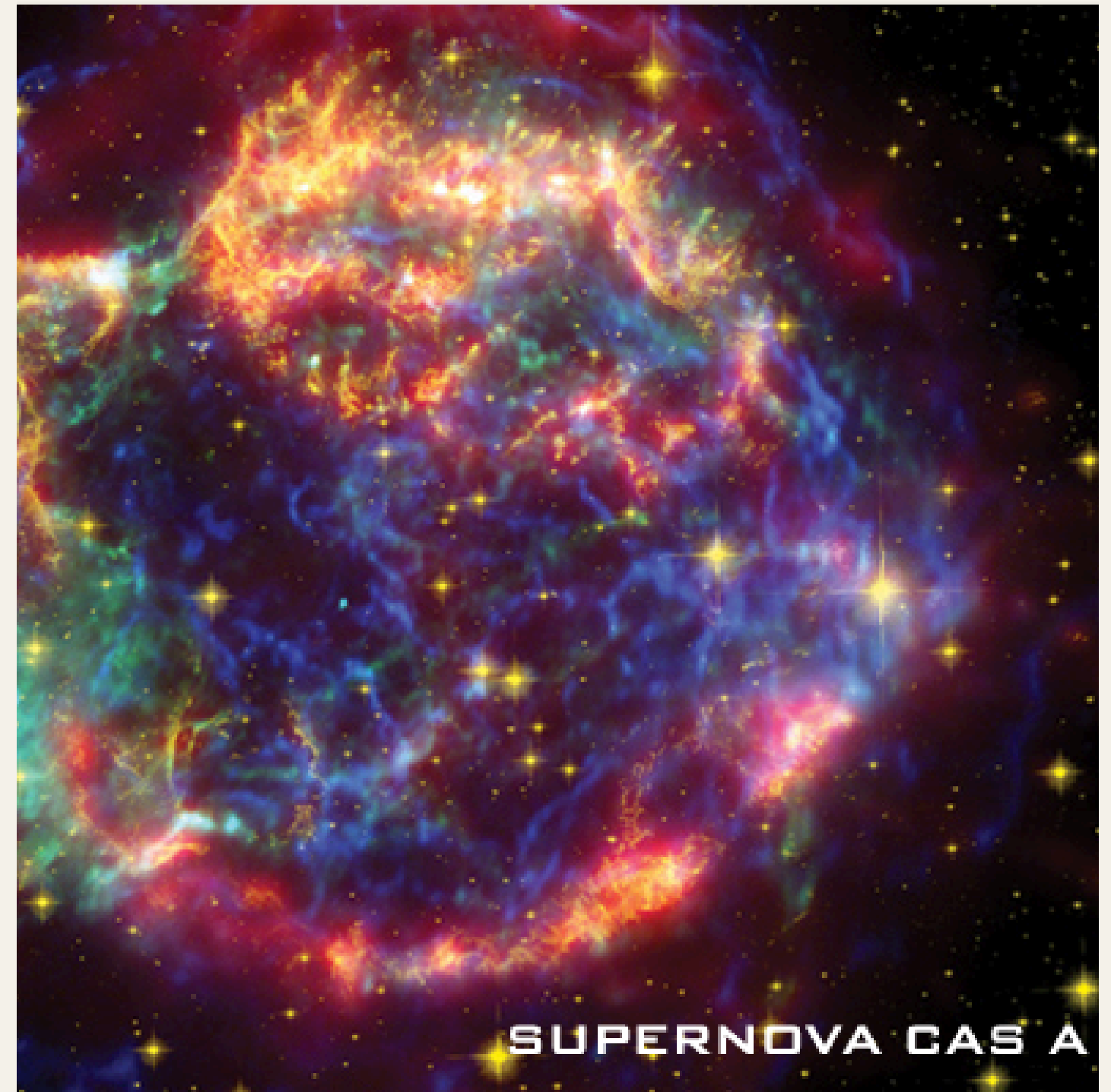




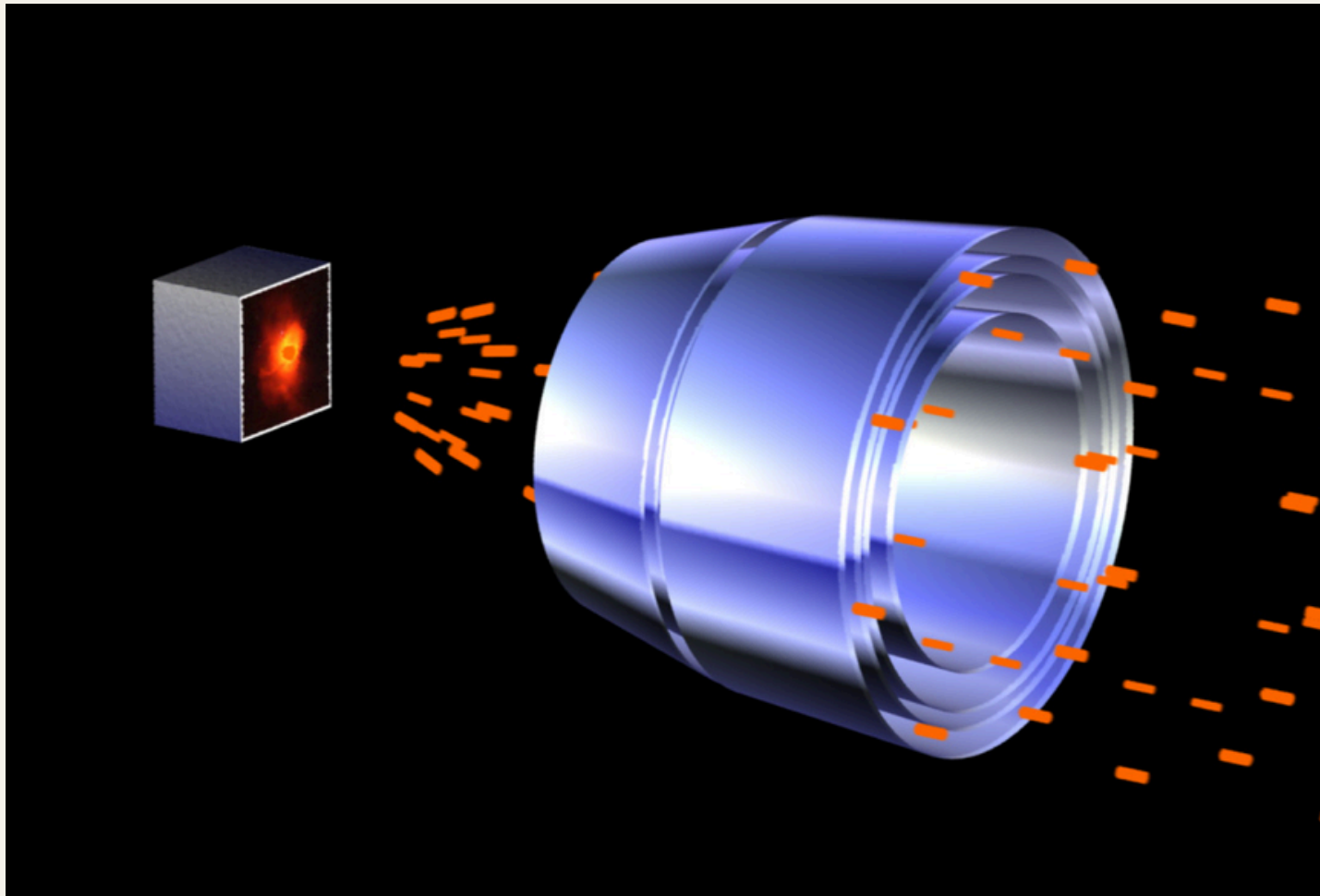
Neil Gehrels Swift Observatory

Why Study High-Energy Phenomena?

- X-rays are a high-energy portion of the electromagnetic spectrum, with photon energies typically ranging from 0.1 to 100 keV.
- Their production is associated with extreme cosmic phenomena: provides insights into astrophysical processes under conditions of high temperature, density, and energy, which are not replicable on Earth.
- X-ray observations are crucial for studying various astrophysical phenomena, including: Black Holes, Supernova Remnants, Galaxy Clusters, Neutron Stars and White Dwarfs, Stellar Coronas and Flares.
- X-ray astronomy serves as a tool for investigating the most energetic and exotic cosmic phenomena.



X-ray Telescopes



X-rays behave differently than visible light. They don't reflect easily off mirrors like regular light.

If an X-ray hits a mirror head-on (at 90 degrees), it's either absorbed or passes right through. No reflection!

The Solution: Grazing Angles!

To reflect X-rays, we have to make them "skim" off the mirror surface at extremely shallow, almost parallel angles.

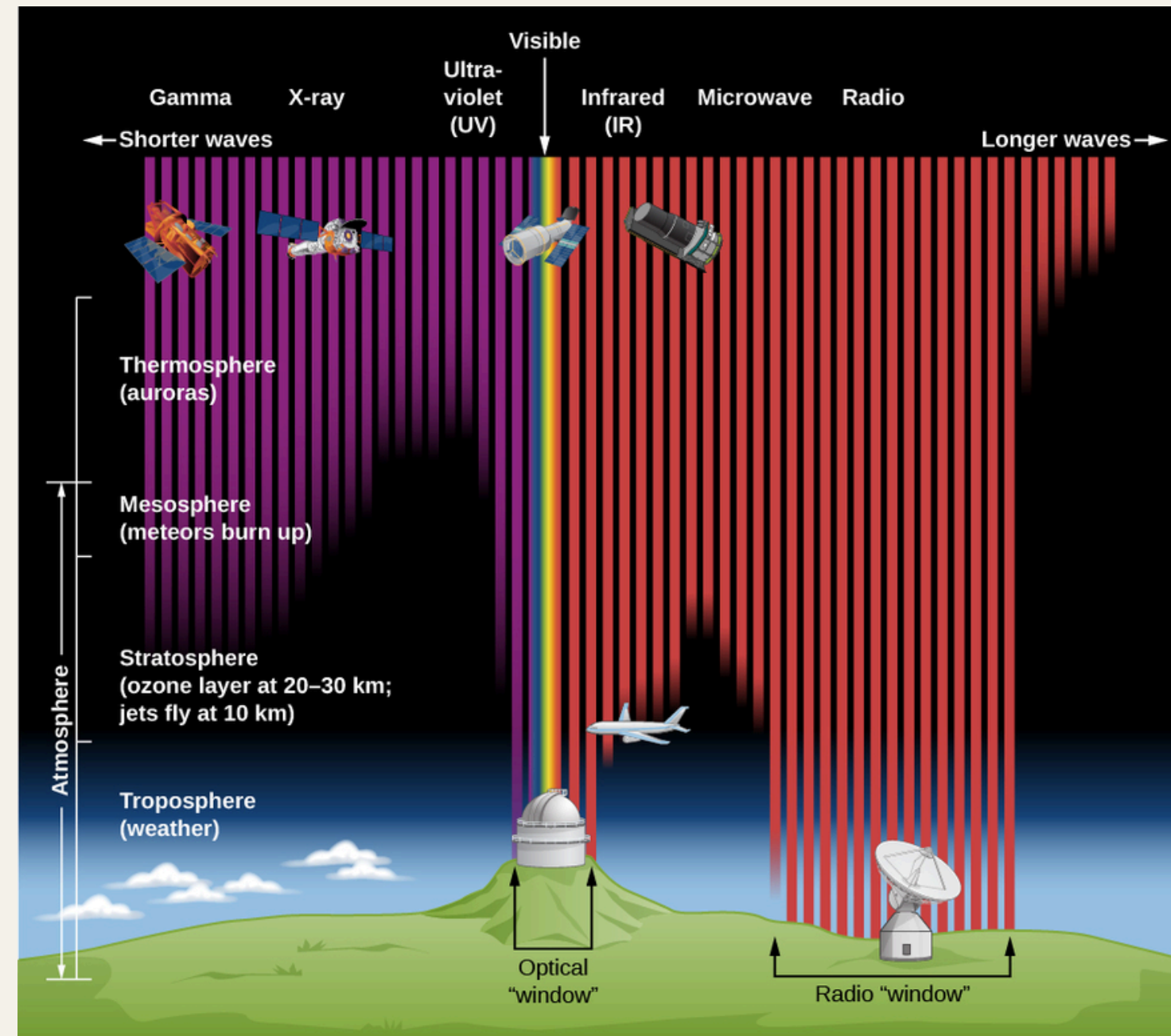
The "***Total External Reflection***" allows X-rays to bounce off incredibly smooth metal surfaces (like gold, platinum, or iridium) when they hit at these very shallow angles.

***Total External Reflection (TER)** is a phenomenon that occurs when X-rays or other very short wavelength electromagnetic waves hit a surface at a very small grazing angle, and instead of being absorbed or transmitted into the material, they are completely reflected.*

Once X-rays are focused using grazing-incidence mirrors, they are recorded by X-ray detectors positioned at the telescope's focal plane. These detectors are designed to measure individual X-ray photons with high precision.

Charge-Coupled Devices (CCDs) are silicon-based semiconductor detectors widely used in soft X-ray astronomy, particularly in the 0.1 to 10 keV energy range. When an incoming X-ray photon strikes the CCD, it deposits energy into the silicon, generating a number of electron-hole pairs proportional to the photon's energy. These charges are collected and stored in discrete potential wells that form a pixel array. After exposure, the charges are shifted across the chip and read out sequentially. This process enables precise measurements of both the position and energy of each photon, allowing for high-resolution imaging and moderate spectral resolution. To reduce thermal noise and improve sensitivity, CCDs are typically cooled to very low temperatures.

Cadmium Zinc Telluride (CZT) arrays, on the other hand, are solid-state detectors optimized for higher-energy X-rays, typically up to around 150 keV. When a high-energy photon interacts with the CZT crystal, it generates electron-hole pairs that are collected by an array of electrodes. The resulting signal provides information about the photon's energy, and in pixelated designs, its approximate position as well.



Earth's atmosphere effectively absorbs most incoming X-rays and gamma-rays from space. Due to this absorption, conducting X-ray astronomy from the ground is impossible. To overcome this limitation, X-ray telescopes must be positioned above the atmosphere, typically using high-altitude balloons or, more effectively, spacecraft.



Neil Gehrels Swift Observatory

Neil Gehrels Swift Observatory is a satellite that studies gamma-ray bursts, the most powerful explosions in the universe, and other cosmic objects and events. Swift houses three multiwavelength telescopes, collecting data in visible, ultraviolet, X-ray, and gamma-ray light.

Swift was launched on 20 November 2004, aboard a Delta II launch vehicle.

United States (NASA), United Kingdom (UK Space Agency), Italy (Italian Space Agency - ASI)

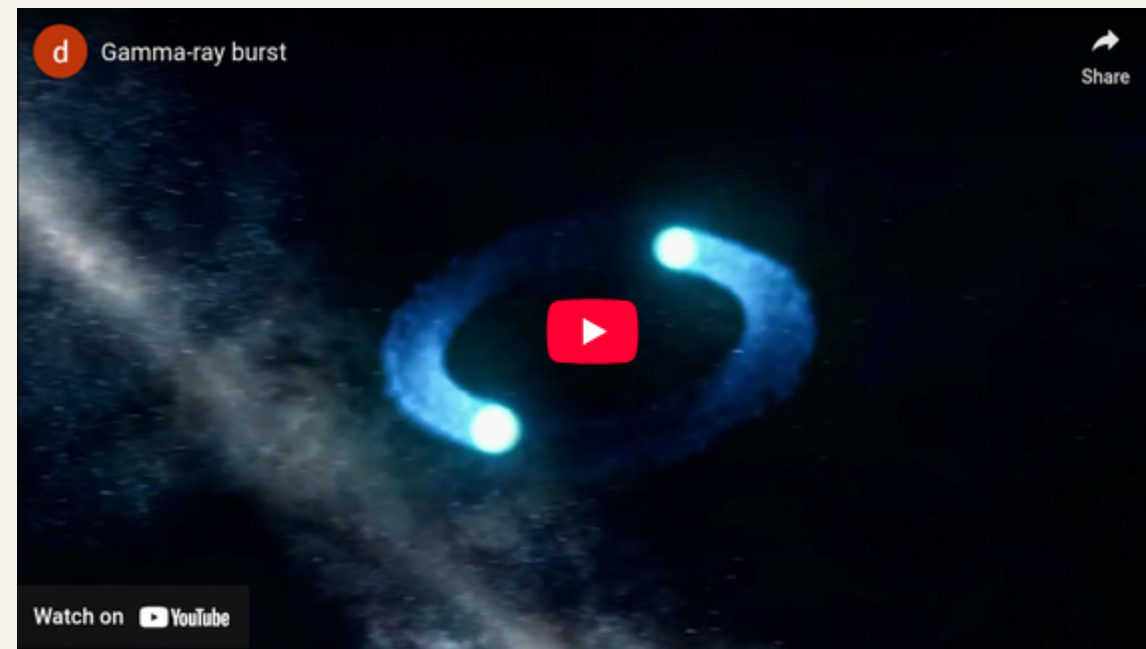
What Are Gamma-Ray Bursts (GRBs)?



- Gamma-Ray Bursts are extremely energetic explosions observed in distant galaxies.
- They are the brightest electromagnetic events known to occur in the universe.
- GRBs typically last from milliseconds to several minutes, followed by an afterglow in lower-energy wavelengths (X-ray, optical, radio).
- They are believed to result from cataclysmic cosmic events, such as:
 - The core collapse of massive stars
 - The merger of compact objects
- GRBs provide critical insights into high-energy astrophysics, stellar evolution, and the early universe.



<https://www.youtube.com/watch?v=nwZSO6ULI2o>



<https://www.youtube.com/watch?v=KofGuzKeJcI>

Swift operates in a Low-Earth Orbit (LEO) at an approximate altitude of 400-600 km, with an orbital inclination of 20.6 degrees. This orbital configuration results in the observatory completing roughly 15.37 revolutions per day.

Swift employs a dual-path system for uplink and downlink of data, designed to ensure both rapid alerts and comprehensive data transfer:

Real-time Alerts: Critical GRB alerts and their initial characteristics are relayed almost instantaneously (at 2 kbps) via the NASA TDRSS (Tracking and Data Relay Satellite System) space data link to the GRB Coordinate Network (GCN). This rapid distribution, often triggering alerts for astronomers worldwide, is vital. This immediate dissemination allows other ground-based and space-based observatories to quickly re-point their instruments and initiate follow-up observations of the same transient event, forming a coordinated multi-observatory campaign.

Stored Data: A higher data rate (2.25 Mbps) is utilized for the bulk of the stored scientific data, which is primarily transmitted via a "store-and-dump" method through the Malindi ground station.

Once on the ground, the Swift Science Data Center (SDC) and its primary archive are located at the Goddard Space Flight Center (GSFC) in the United States. Concurrently, the UK Swift Science Data Centre is hosted at the University of Leicester. These centers are responsible for the processing, archiving, and distribution of the mission's extensive dataset.

Mission Operations: Autonomous Triggers and Guest Investigator Program

Swift's core operational mode for GRBs is characterized by a high degree of autonomy. The Burst Alert Telescope (BAT) detects a gamma-ray burst, calculates a coarse position, and then autonomously triggers the spacecraft to slew. This re-points the narrow-field XRT and UVOT instruments to the burst location within seconds.

Beyond its dedicated GRB follow-up, Swift allocates a substantial portion of its observing time to the Guest Investigator (GI) and Target of Opportunity (ToO) programs.

Guest Investigator (GI) Program: A substantial portion of Swift's observing time is allocated to the GI program. Scientists from around the world can submit proposals to use Swift's instruments (BAT, XRT, and UVOT) to observe a wide variety of astrophysical phenomena. Blazars are frequently proposed targets due to their highly variable nature and broad-band emission. These proposals are peer-reviewed, and successful ones are scheduled during the periods when Swift isn't actively responding to a GRB.

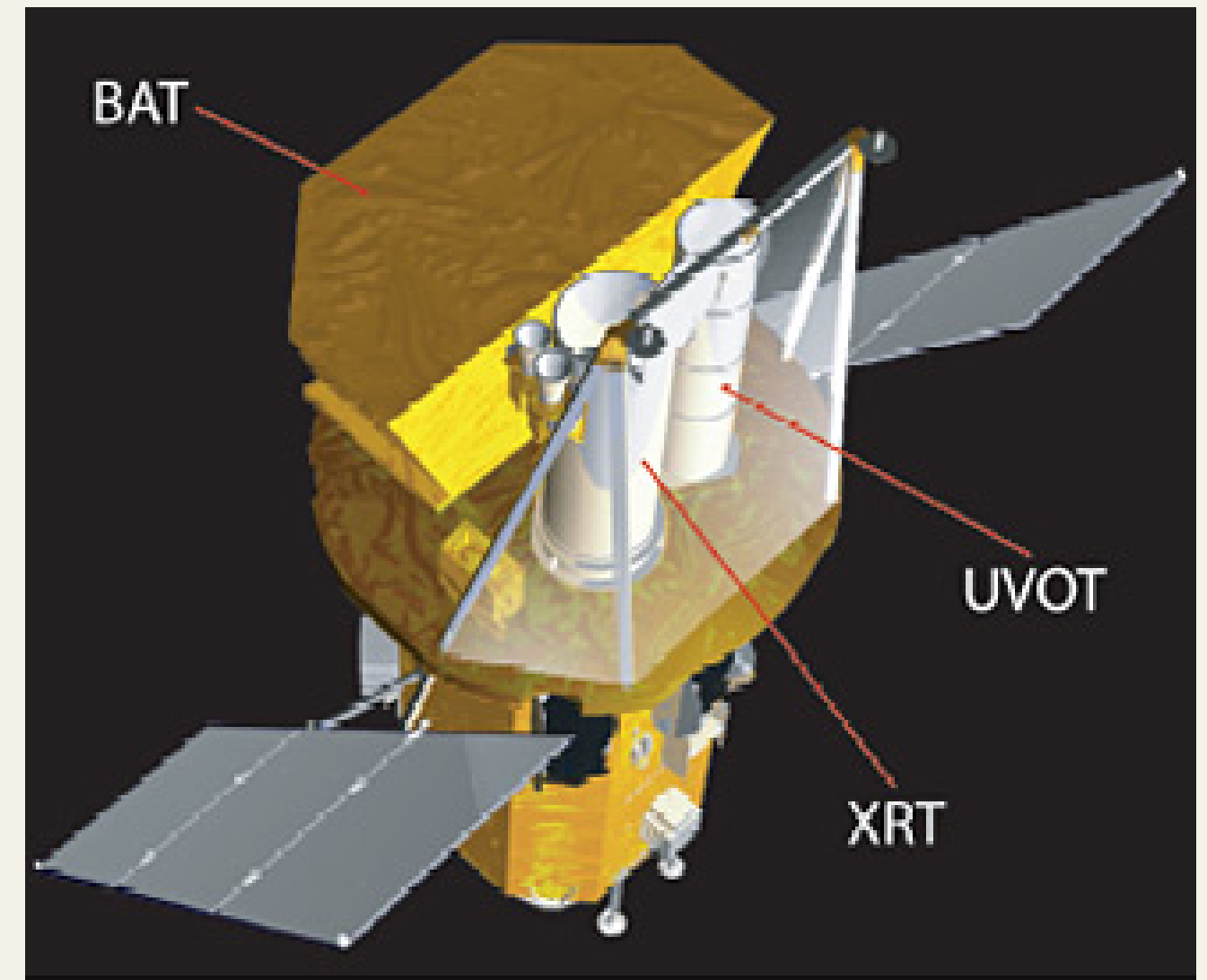
Target of Opportunity (ToO) Program: Blazars are known for their extreme and rapid variability across the electromagnetic spectrum, often undergoing dramatic flares. When a blazar undergoes a significant outburst (e.g., in gamma-rays detected by Fermi-LAT, or at other wavelengths), scientists can submit a ToO request to Swift. This triggers rapid observations of the flaring blazar, allowing Swift's XRT and UVOT to capture the evolving spectral and temporal characteristics of the flare.

Swift's Instruments: BAT

The **Burst Alert Telescope (BAT)** serves as Swift's primary instrument for detecting gamma-ray bursts and providing their initial positions. It is a coded aperture imaging instrument with a wide field-of-view (FOV) of 1.4 steradians (half-coded), enabling it to monitor approximately one-sixth of the entire sky at any given time. The BAT's operational energy range is 15-150 keV (imaging), up to 500 keV (non-coded)

The BAT operates in two distinct modes: a burst mode for detecting GRB triggers and a survey mode for collecting hard X-ray survey data.

In burst mode, a trigger algorithm continuously monitors for excesses in the detector count rate above background, applying a variety of criteria to identify genuine bursts. Upon detection, the BAT swiftly calculates an initial position (with an accuracy of 1-4 arcminutes) within seconds and, if warranted, sends a command for the spacecraft to slew to the burst location. This decision is made by a "figure-of-merit (FOM)" algorithm embedded in the flight software, which prioritizes new, interesting transients.

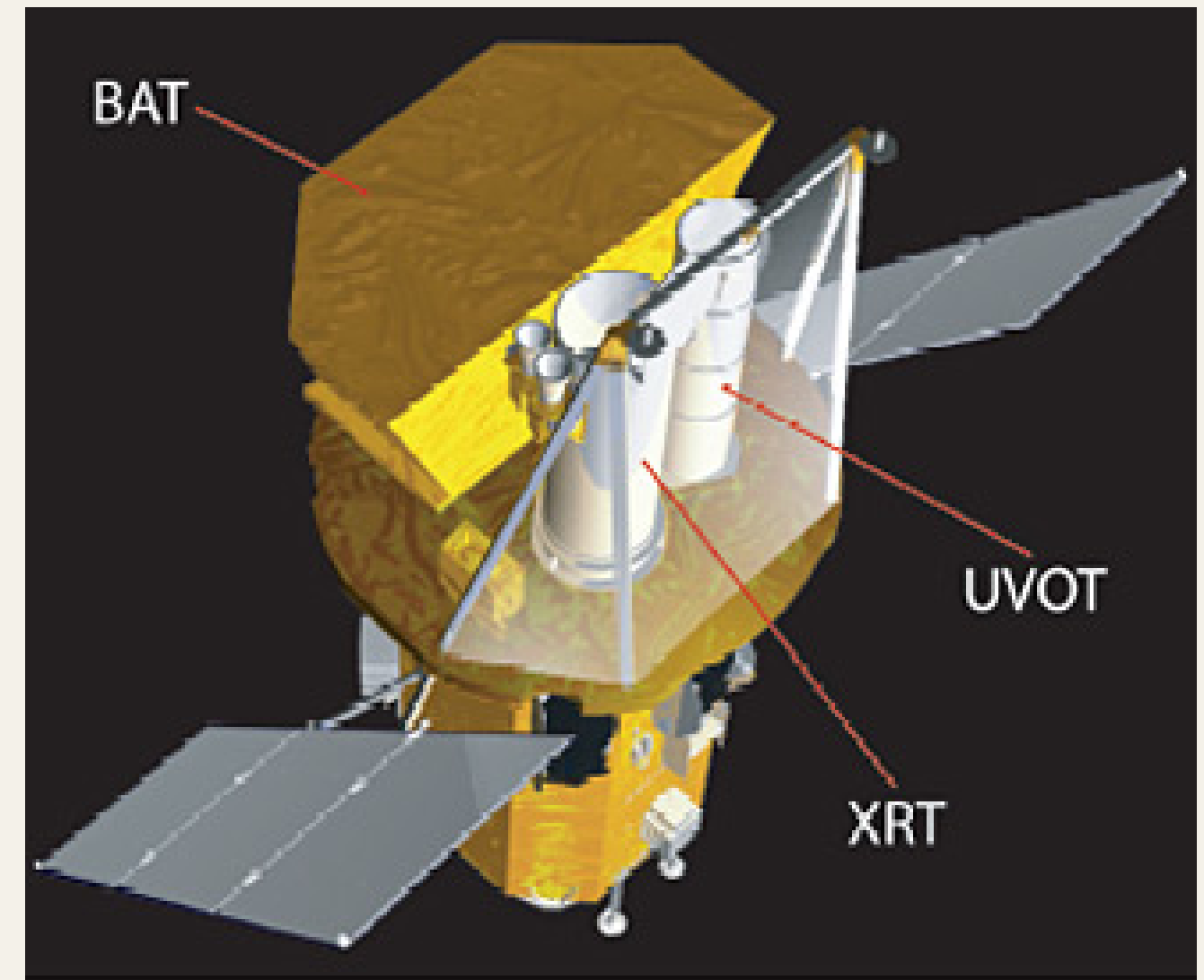


Swift's Instruments: XRT

The **X-ray Telescope (XRT)** is a focusing X-ray instrument specifically designed to measure the fluxes, spectra, and light curves of GRBs and their afterglows across an exceptionally wide dynamic range, spanning more than seven orders of magnitude in flux. It is crucial for providing high-precision X-ray positions and detailed spectra of these events.

The XRT employs a grazing incidence Wolter-I telescope to focus X-rays onto a state-of-the-art CCD detector.

The XRT's operational energy range is 0.2-10 keV. Its field of view (FOV) is 23.6 x 23.6 arcminutes.



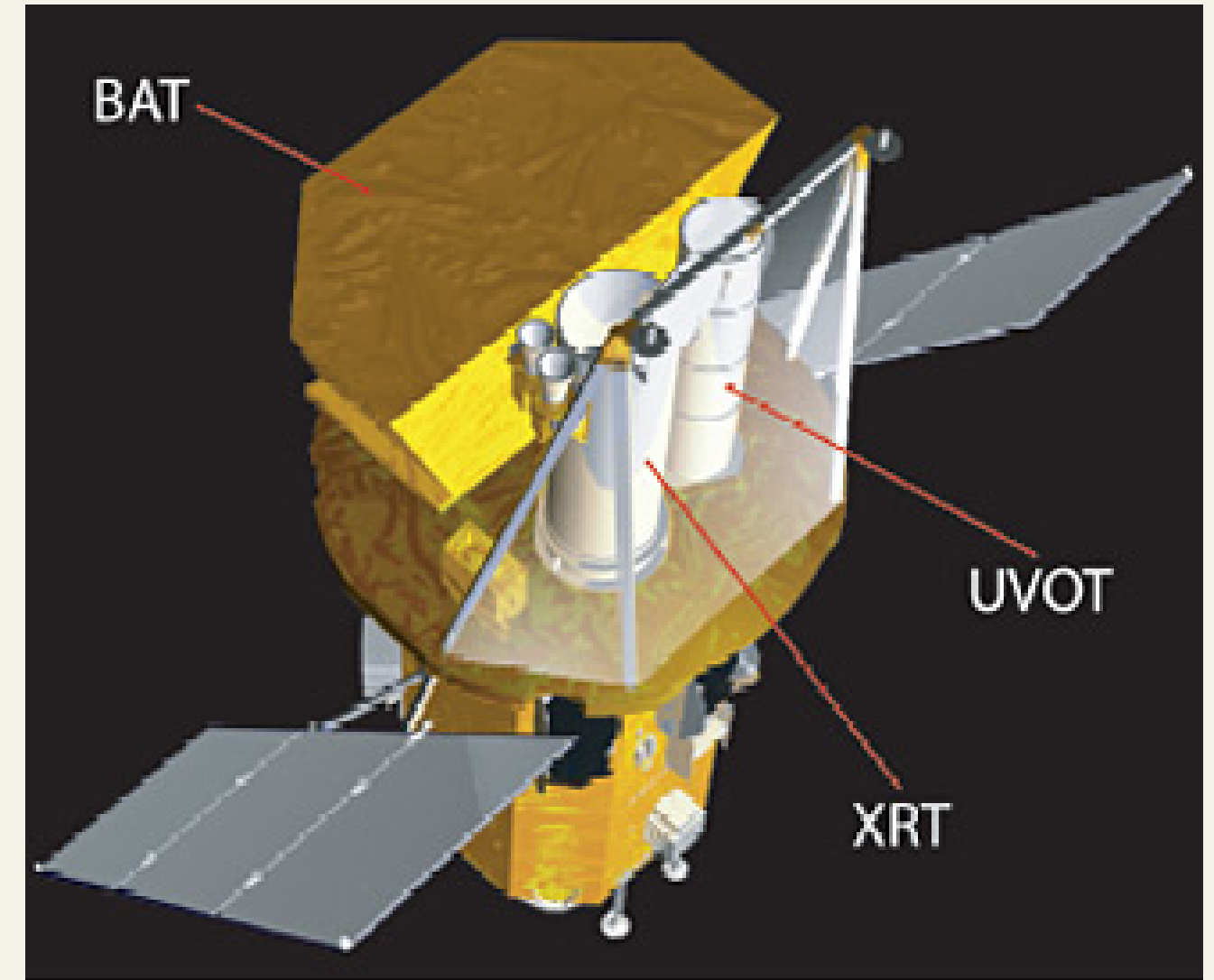
Swift's Instruments: XRT

The XRT supports multiple readout modes, autonomously switching between them based on the source flux to optimize data collection for the highly dynamic GRB afterglows.

Photon-counting (PC) Mode: This mode provides full spectral and spatial information, suitable for lower flux sources.

Windowed Timing (WT) Mode: This mode sacrifices some spatial information to achieve higher time resolution, crucial for very bright, rapidly varying sources.

Photodiode (PD) mode: Offers very high time resolution but no spatial information. (no longer operational due to technical and operational challenges - it was disabled)



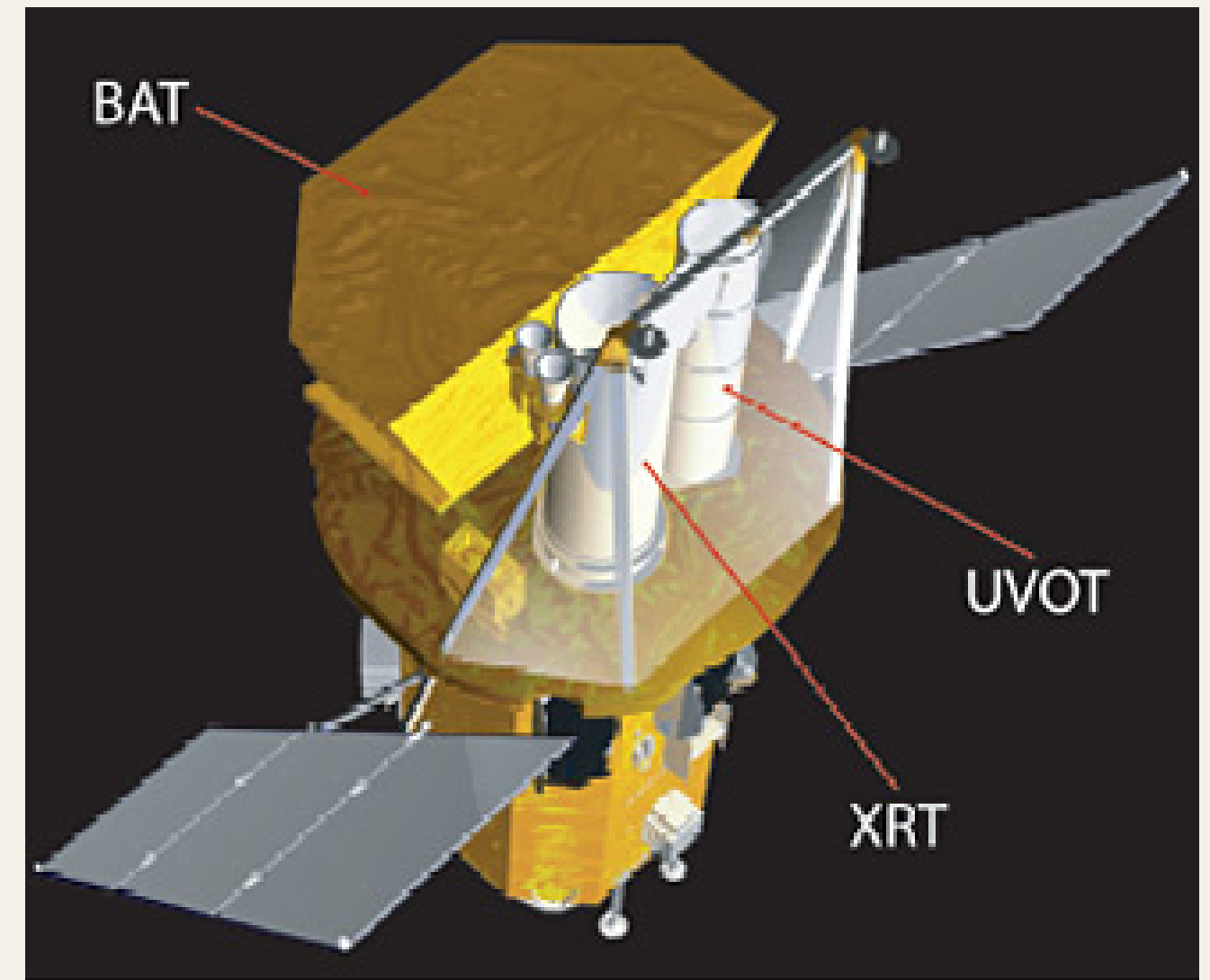
Swift's Instruments: UVOT

The **Ultraviolet/Optical Telescope (UVOT)** completes Swift's multi-wavelength observational suite by providing simultaneous ultraviolet and optical coverage. It plays a critical role in imaging the GRB field and transmitting finding charts to ground-based observatories.

The UVOT is a 30 cm modified Ritchey-Chrétien UV/optical telescope, precisely co-aligned with the XRT.

The instrument covers a broad wavelength range from 170 to 650 nm. Its field of view (FOV) is 17×17 arcminutes, with an angular resolution of 0.9 arcsec.

The UVOT's capabilities are further enhanced by an 11-position filter wheel, which facilitates broad-band UV/visible photometry using filters such as U, B, V, UVW1, UVM2, UVW2, and a White light filter. It also offers low-resolution grism spectroscopy (UV grism, Optical grism) for brighter GRBs.



Swift Data Processing Pipeline and Data Levels

Level 0: These are the raw telemetry files directly downlinked from the spacecraft. They represent the fundamental, unprocessed data stream from the instruments.

Level 1: At this stage, raw telemetry is converted into FITS files, undergoing minimal initial processing. For instance, UVOT raw images are provided at this level (`sw[obsid]u[filter]_rw.img.gz`). These files serve as the initial structured data for subsequent instrument-specific processing.

Level 2: This level comprises calibrated event lists and cleaned sky images, produced after rigorous data screening and coordinate transformations. Most users typically commence their scientific analysis with these files. Examples include cleaned XRT event files (`sw[obsid]xpc_cl.evt.gz`) and UVOT sky coordinate images (`sw[obsid]u[filter]_sk.img.gz`). These data are corrected for instrumental effects and mapped to celestial coordinates, making them ready for scientific interpretation.

Level 3: These are higher-level science products, automatically generated from the Level 2 files. This includes products such as light curves (`sw[obsid]xpc_lc.fits`) and spectra (`sw[obsid]xpc_spec.fits`). These products are directly usable for scientific interpretation and publication.

